Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *MENTALCLOUDS* | ***Context:*** *South Asia and MENA NLP/AI Practitioner Cohort — MentalCLOUDS Assessment*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Automatic evaluation uses ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore [Q20, Q21], all computed against English reference summaries via 'overlap of n-grams... between the generated summary and reference summaries' [Q22, Q23] and 'longest co-occurring n-gram' [Q24]. For the HIGH-priority OF dimension, this design provides zero valid evaluation signal for systems generating summaries in Arabic, Hindi, Urdu, Bengali, Farsi, or Dari — covering the majority of MENA-focused practitioners and many South Asian sub-populations. The 2026 multilingual mental health evaluation study confirms that for Arabic, BLEU scores ~0.54 vs LaBSE ~0.78 demonstrate n-gram metrics are 'especially poorly suited for Arabic clinical NLP evaluation' [WEB-8]. The XLingEval benchmark documents statistically significant English-vs-Hindi BERTScore performance gaps confirming validity concerns for non-English clinical NLP [WEB-10]. AraHealthQA 2025 explicitly found BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA [WEB-12]. Cross-lingual alternatives (LaBSE, XLM-RoBERTa BERTScore, Arabic-specific BERTScore via AraBERT/CAMeLBERT/MARBERT, Hindi/Urdu via MuRIL/IndicBERT) are technically viable [WEB-8, WEB-9, WEB-11] but have not been applied to MentalCLOUDS or to clinical summarization in target languages. The qualitative six-parameter expert evaluation [Q45] is partially region-transferable as scaffolding but inherits the calibration concerns documented in OO. Per A4, the benchmark 'is applicable only to English-language systems; multilingual evaluation requires an entirely separate framework.'

ID	Evidence
Q20	“Given the generative nature of the task, we employ standard summarization evaluation metrics such as Rouge-1 (R-1), Rouge-2 (R-2), Rouge-L (R-L), and BERTScore (BS) along with their corresponding Precision (P), Recall (R) and F1 scores.” (p.9)
Q21	“Since F1 accounts for Precision and Recall, we compare LLM's performance based on F1 unless stated otherwise.” (p.9)
Q22	“ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [58] assesses the overlap of n-grams (sequences of n consecutive words) between the generated summary and reference summaries.” (p.9)
Q23	“This metric measures the number of overlapping units such as n-gram, word sequences, and word pairs between the generated summary evaluated against the gold summary typically created by humans.” (p.9)
Q24	“ROUGE-L takes into account the longest co-occurring n-gram between the candidate and the reference summaries.” (p.9)
Q45	“Experts evaluate each summary against these acceptability parameters, assigning continuous ratings on a scale from 0 to 2, where a higher rating signifies enhanced acceptability.” (p.12)
WEB-8	arXiv 2602.02440 — Arabic LLM evaluation: BLEU ~0.54 vs LaBSE ~0.78 confirms n-gram metrics like ROUGE poorly suited for Arabic clinical NLP (https://arxiv.org/html/2602.02440)
WEB-9	MDPI Healthcare 2025 — AraBERT, CAMeLBERT, MARBERT applied to Arabic mental health classification; BERTScore for Arabic clinical summarization not yet evaluated (https://www.mdpi.com/2227-9032/13/9/985)
WEB-10	ACM Web Conference 2024 (XLingEval) — statistically significant BERTScore gap between English (0.9206) and Hindi for healthcare queries, confirming metric validity concerns for non-English clinical NLP (https://dl.acm.org/doi/10.1145/3589334.3645643)
WEB-11	MDPI Applied Sciences 2025 — XLM-RoBERTa BERTScore demonstrated robust for cross-lingual summarization including low-resource languages; not yet applied to clinical mental health summarization (https://www.mdpi.com/2076-3417/15/14/7800)
WEB-12	AraHealthQA 2025 — BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA (https://arxiv.org/html/2508.20047v2)

Strengths

- For English-system deployment, ROUGE/BERTScore metrics are standard summarization metrics with established interpretability [Q20, Q21].
- Dual quantitative + qualitative pipeline [Q5, Q18, Q25] reduces over-reliance on surface-overlap metrics for English deployment.
- Hallucination scale [Q46] and per-conversation reporting (39 test conversations) [Q53, Q55] provide a partly language-agnostic safety signal.

ID	Evidence
Q5	"The generated summaries are evaluated quantitatively using standard summarization metrics and verified qualitatively by mental health professionals." (p.1)
Q18	"We undertake a comprehensive evaluation of the generated summaries across various architectures, employing a dual approach of quantitative and qualitative assessments." (p.9)
Q20	"Given the generative nature of the task, we employ standard summarization evaluation metrics such as Rouge-1 (R-1), Rouge-2 (R-2), Rouge-L (R-L), and BERTScore (BS) along with their corresponding Precision (P), Recall (R) and F1 scores." (p.9)
Q21	"Since F1 accounts for Precision and Recall, we compare LLM's performance based on F1 unless stated otherwise." (p.9)
Q25	"Notably, in the context of counseling summaries, which are inherently tied to a domain-specific conversation, we embark on a meticulous qualitative examination of the generated summaries for individual counseling components." (p.10)
Q46	"Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed)." (p.12)
Q53	"Additionally, the evaluation of hallucination identification is divided into three categories: no hallucination observed, hallucination barely observed, and too much hallucinated in a set of 39 conversations." (p.14)
Q55	"Out of the 39 test conversations, the majority of cases on average demonstrate no hallucination, with Mistral and MentalBART achieving 75.13% and 76.15% each, while MentalLlama shows a slightly higher value of 77.69%." (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (free-text summaries) matches deployment for English systems but provides zero valid signal for Arabic, Hindi, Urdu, Bengali, Farsi, or Dari output [Q22, Q23, WEB-8, WEB-10, WEB-12].

OF-2: INSUFFICIENT DOCUMENTATION — text-to-speech is not within the benchmark's scope. Counseling deployment in low-literacy regional sub-populations may require speech output, but the benchmark does not address this.

OF-3: Literacy and accessibility considerations are not addressed in the benchmark documentation. Regional variation in literacy rates and clinical-language register (e.g., Arabic diglossia between dialect and MSA) are not modeled in the output form.

OF-4: Documented form mismatches harming external validity: (a) ROUGE/BERTScore against English-only references provide no valid signal for non-English output [Q22, Q23, WEB-8]; (b) no multilingual reference summaries provided; (c) no cross-lingual metric variants documented; (d) qualitative parameter calibration carries over OO concerns; (e) cross-lingual alternatives (LaBSE, XLM-R BERTScore, AraBERT-based BERTScore) viable but unapplied [WEB-8, WEB-9, WEB-11].

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Q22	“ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [58] assesses the overlap of n-grams (sequences of n consecutive words) between the generated summary and reference summaries.” (p.9)
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Information Gaps

- No documented exploration of multilingual BERTScore variants or cross-lingual evaluation pathways within the benchmark itself.
- No reported behavior of the benchmark when evaluating machine-translated outputs.

Requires Expert Verification

- Cross-lingual NLP evaluators should pilot LaBSE and XLM-R BERTScore on a translated MentalCLOUDS reference subset to characterize the magnitude of cross-lingual signal loss empirically.

Remediation

Gap 1: ROUGE and BERTScore against English-only references [Q22, Q23] provide no valid signal for Arabic, Hindi, Urdu, Bengali, Farsi, or Dari output [WEB-8, WEB-10, WEB-12].

Recommendation: Pilot cross-lingual metrics (LaBSE [WEB-8], XLM-R BERTScore [WEB-11], AraBERT/CAMeLBERT/MARBERT-based BERTScore for Arabic [WEB-9], MuRIL/IndicBERT for Hindi/Urdu) on a translated reference subset and validate against expert ratings; do not report ROUGE for non-English output systems.

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